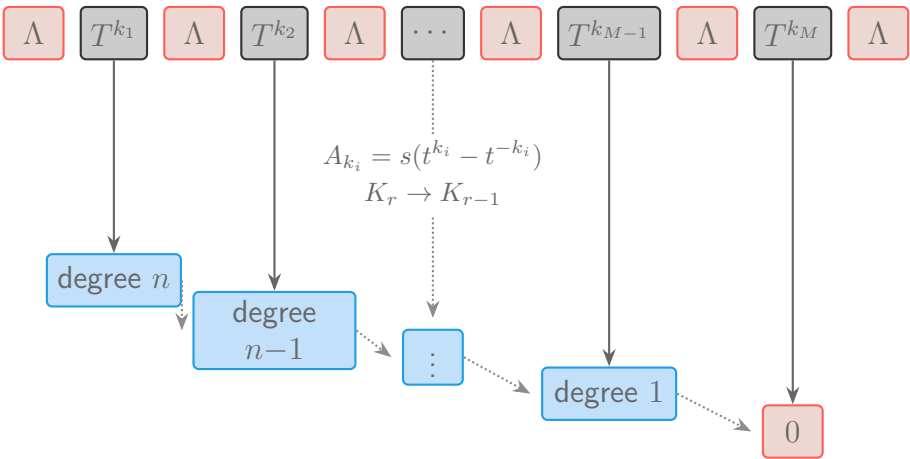


$$n = N/2$$



$$M \geq N/2 \Rightarrow \text{word} = 0$$